

Assignment 4 - Practice Quiz A

20 questions; 30 minutes suggested. Practice does not affect your grade.

1. What is the LCS length of "banana" and "ananas"?

Answer: _____

2. For $A = \text{"bcaba"}$ and $B = \text{"caaa"}$, $dp[3][2]$ uses prefix-length indexing. What does it represent?

- A. The number of all common subsequences.
- B. LCS length of "ba" and "aa".
- C. LCS length of "bca" and "ca".
- D. Longest common substring length of "bca" and "ca".

3. $A = \text{"gceihdjb"}$, $B = \text{"gehj"}$. A flawed implementation computes longest common SUBSTRING instead of SUBSEQUENCE. By how much is its answer length smaller than the LCS length?

Answer: _____

4. Adjacent archive blocks have sizes [8, 9, 8, 8]. Each merge costs the sum of the two merged block sizes. What is the minimum total cost?

Answer: _____

5. Blocks [6, 8, 10]. First merge original blocks 0 and 1; then merge their result with block 2. What is the total cost of this plan?

Answer: _____

6. A single archive block has size 600000000. What is its minimum merge cost?

Answer: _____

7. Tree weights by vertex ID: [0, 6, 6, 8, 6]; edges [(0, 1), (0, 2), (2, 3), (2, 4)]. The empty selection is allowed. What is the maximum total weight with no adjacent selected vertices?

Answer: _____

8. Node u has two children. Their (take, skip) states are (7, 6) and (9, 10). What is $skip[u]$?

Answer: _____

9. Node u has weight 9, two children with skip values [6, 7], and arbitrary take values. What is $take[u]$?

Answer: _____

10. Text="aaaaaaaaaaaa", pattern="aaaaaa". How many occurrences are there when overlaps count?

Answer: _____

11. For KMP pattern "ababbab", what is the last prefix-function value (longest proper prefix also a suffix)?

Answer: _____

12. Base=31, modulus=101, ASCII codes. The two-character strings "af" and "dn" have equal hashes. Text="dnafafafafdn"; pattern="af". How many occurrences must EXACT Rabin-Karp report?

Answer: _____

13. After KMP fully matches pattern "ababababababab", it sets $j = \pi[m-1]$ to keep overlaps. What is the new j?

Answer: _____

14. Build the suffix array of "bbaab" using exactly its nonempty suffixes and zero-based positions, without a sentinel. What is sa[0]?

Answer: _____

15. Text="abacaba". Overlapping repeated substrings are allowed. What is the longest repeated substring length?

Answer: _____

16. Kasai has just processed a non-first suffix with running common-prefix length $h=8$. The next suffix in text order also has rank greater than zero. Before extending that comparison, what reuse value is kept?

Answer: _____

17. Consider binary searching a suffix array for a length-m pattern, using $O(m)$ character comparisons at each search step. Which is the tightest worst-case bound among these options?

- A. $O(\log \log n)$
- B. $O(m \log n)$
- C. $O(1)$
- D. $O(n \text{ factorial})$

18. Alarm dictionary is ["aaaaaa","aaaaaa","b"]. Text="aaaaaaaaaaaa". Each dictionary position has its own count. What is the count for dictionary position 1?

Answer: _____

19. Patterns are ["he","she","hers"]. Scan "hehe" with Aho-Corasick. How many total occurrences of "he" are counted?

Answer: _____

20. An Aho-Corasick dictionary ["ab","b"] is reused. Query "abababababab" twice as two independent texts. How many occurrences of "ab" should the SECOND query return?

Answer: _____